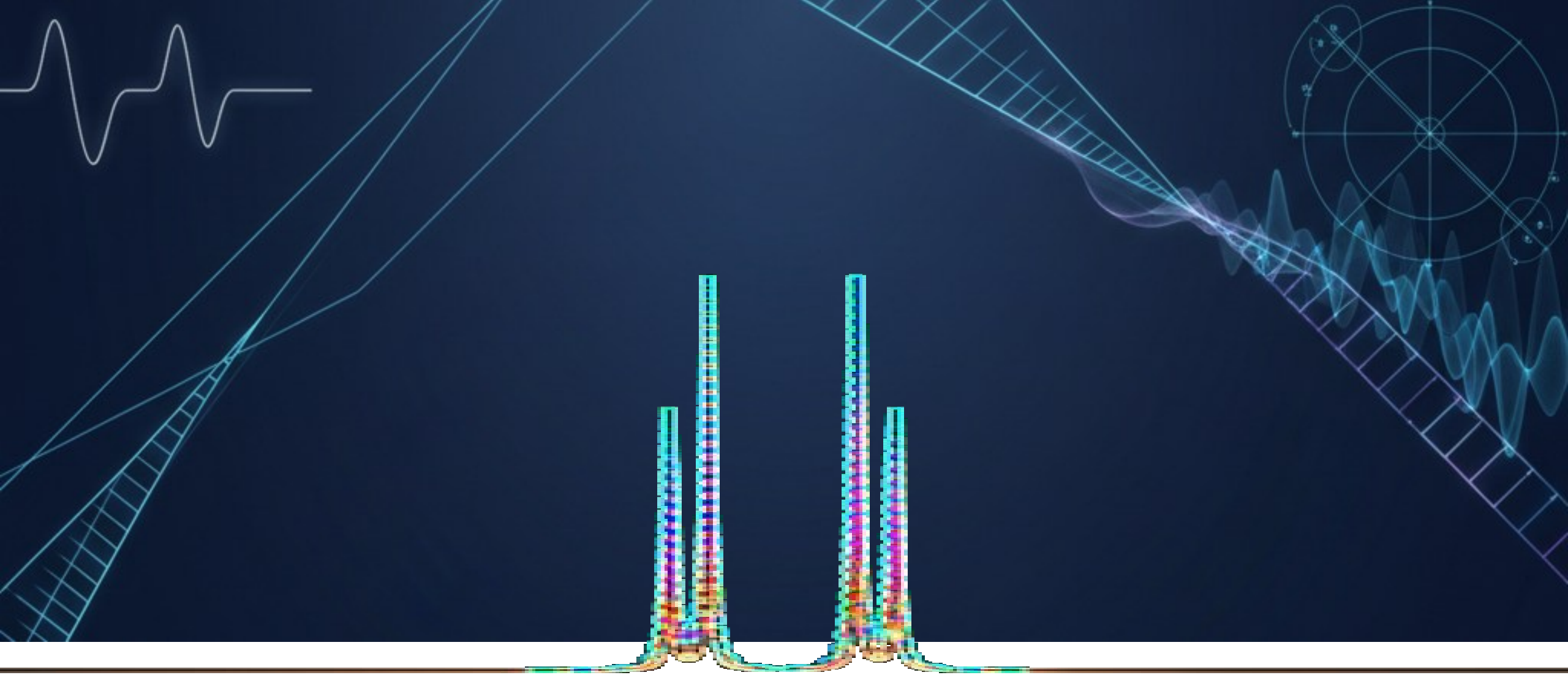


Predicting NMR spectra on a quantum computer

Qiskit Fall Fest 2025 Stuttgart | Team Annihilation Operators

Clustering of states with a similar spin results in significant speed up

Example:

Clusters for 5 spins

↑↑↑↑↑ weight = 1

↑↓↓↓↓ … weight = 5

↑↑↓↓↓ … weight = 10

↑↑↑↓↓ … weight = 10

↑↑↑↑↓ … weight = 5

↓↓↓↓↓ weight = 1

weight = $\binom{m}{N}$

One cluster – subspace of a fixed total spin value

Pick one random string from each cluster

Compute the autocorrelation function
with weights defined by the number
of different strings in a cluster

Liner scale on the number of samples!!

Clustering the whole Hilbert space

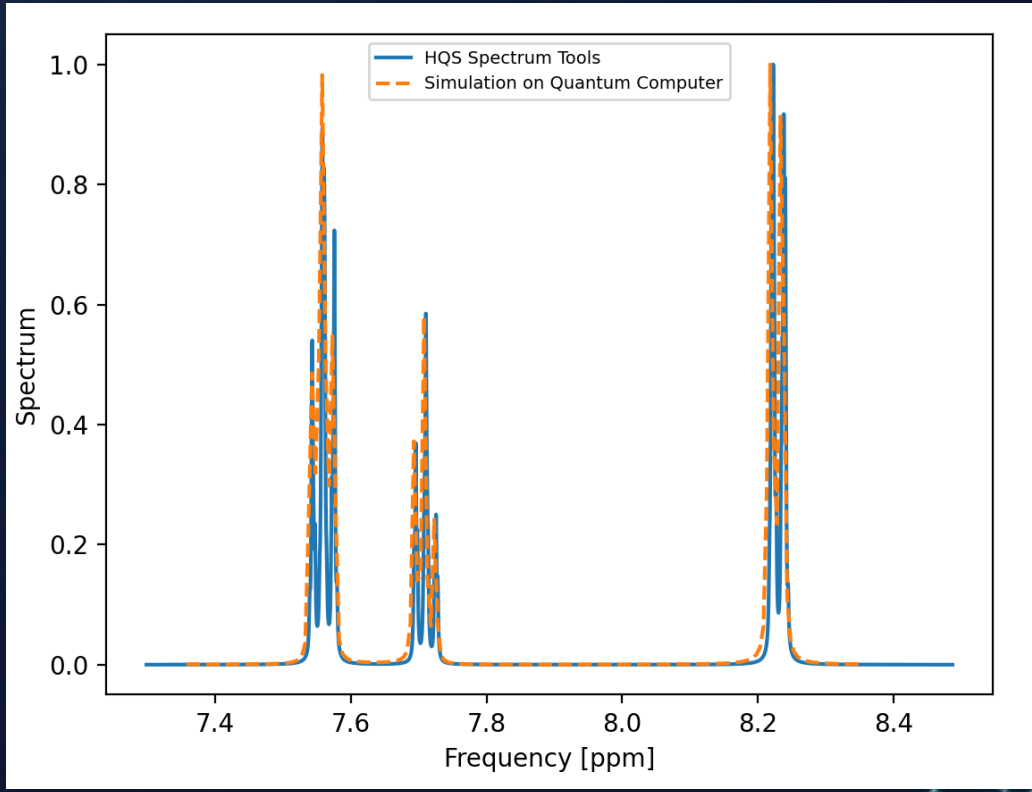
Comparison of our approaches for NMR spectra prediction

- All states of the 5 spin system
- 2^5=32 states (all)
- 8 minutes
- L2 = 0.071

- Only states with > half of spins up
- 16 states (half)
- 4 minutes
- L2=0.071

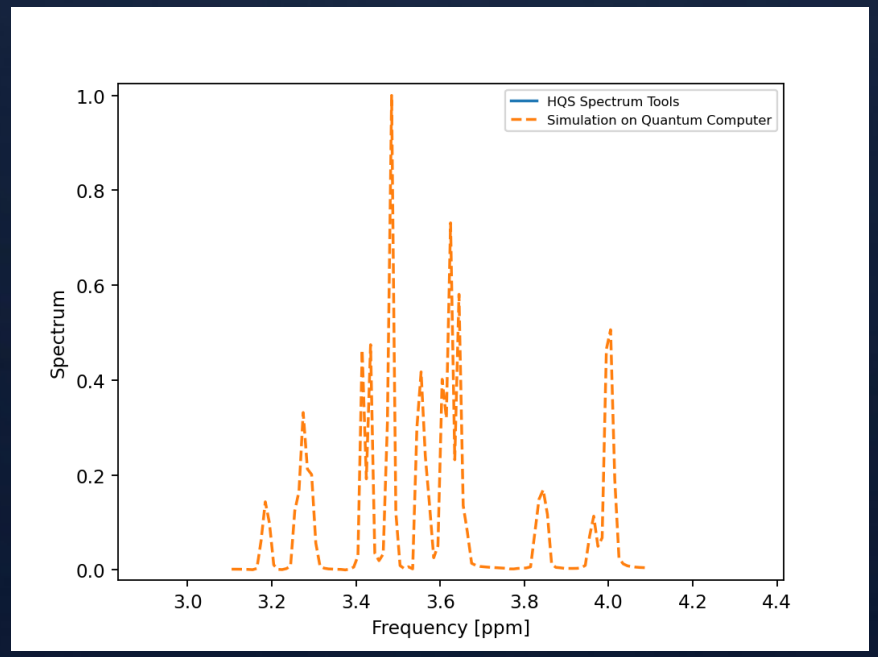
- Cluster states with similar total spin
- 6 states (only N+1)
- 1.5 minutes
- L2 = 0.083

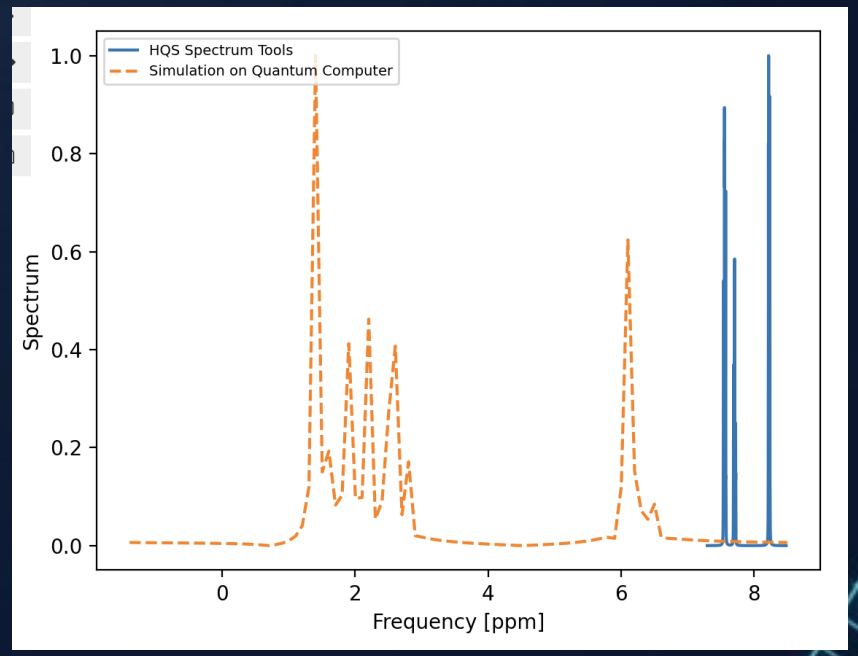
Clustering method, 5 spins, L2=0.083



Runtime comparison of cluster approach for different molecules

- Spins 12
- Run-time 40min
- Evolution time: 0.2





The baseline spectrum for 12 spins coincides with the spectrum for 5 spins

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